

Functional Dependencies

FD Basic definitions

- Let R be a relation, and let X and Y be subsets of set of attribute of R , then Y is functionally dependent on X iff each X value in R has associated with it precisely one Y value in R

FD Basic definitions (cont.)

- Symbol is $X \rightarrow Y$
read X functionally determine Y
or X arrow Y
- X is determinant
- Y is dependent
- $\rightarrow$ is functionally determine

FD Basic definitions (cont.)

Example

<u>S#</u>	CITY	<u>P#</u>	QTY
S1	London	P1	100
S1	London	P2	100
S2	Paris	P1	200
S2	Paris	P2	200
S3	Paris	P2	300
S4	London	P2	400
S4	London	P4	400
S4	London	P5	400

$\{S\# \} \rightarrow \{CITY\}$ or
 $S\# \rightarrow CITY$

FD Basic definitions (cont.)

- other FDs

$\{S\#, P\# \} \rightarrow \{QTY\}$

$\{S\#, P\# \} \rightarrow \{CITY\}$

$\{S\#, P\# \} \rightarrow \{CITY, QTY\}$

$\{S\#, P\# \} \rightarrow \{S\# \}$

$\{S\#, P\# \} \rightarrow \{S\#, P\#, CITY, QTY\}$

$\{S\# \} \rightarrow \{QTY\}$

$\{QTY\} \rightarrow \{S\# \}$

Closure of a set of dependencies

- Simple of the FD

$$\{S\#, P\#\} \rightarrow \{CITY, QTY\}$$

- Implies to

$$\{S\#, P\#\} \rightarrow CITY$$

$$\{S\#, P\#\} \rightarrow QTY$$

- The set of all FDs that are implied by a given set of FDs is called **Closure of FD** or FD^+

Closure of a set of dependencies (cont.)

- Example : $R(A,B,C,D,E,F)$

FD : $A \rightarrow B$

$A \rightarrow C$

$CD \rightarrow E$

$CD \rightarrow F$

$B \rightarrow E$

- Computed FD^+ from FD : $A \rightarrow E$, $A \rightarrow BC$, $CD \rightarrow EF$ etc.
- Let $R(A,B,C,D,E,F)$ and
 $FD = \{ A \rightarrow B, A \rightarrow C, CD \rightarrow E, CD \rightarrow F, B \rightarrow E \}$ then
 $FD^+ = \{ A \rightarrow E, A \rightarrow BC, CD \rightarrow EF, \dots \}$

Closure of a set of dependencies (cont.)

- The algorithm to compute FD^+ from FD is **inference rules** called **Armstrong's axioms**
- Let A , B , and C be subsets of the set of attributes of the given relation R , and AB mean union of A and B then :

Closure of a set of dependencies (cont.)

- 1. Reflexivity** : If B is a subset of A ,
then $A \rightarrow B$
- 2. Augmentation** : If $A \rightarrow B$,
then $AC \rightarrow BC$
- 3. Transitivity** : If $A \rightarrow B$ and $B \rightarrow C$,
then $A \rightarrow C$

Closure of a set of dependencies (cont.)

- Several further rules can be derived from the three given above.

4. Self-determination : $A \rightarrow A$

5. Decomposition : If $A \rightarrow BC$,
then $A \rightarrow B$ and $A \rightarrow C$

6. Union : If $A \rightarrow B$ and $A \rightarrow C$,
then $A \rightarrow BC$

7. Composition : If $A \rightarrow B$ and $C \rightarrow D$,
then $AC \rightarrow BD$,

Closure of a set of dependencies (cont.)

- Darwen proves the following rule, called the **General Unification Theorem** :

8. If $A \rightarrow B$ and $C \rightarrow D$,

then $A \cup (C - B) \rightarrow BD$

where $\cup$ is union and $-$ is set of difference

Closure of a set of dependencies (cont.)

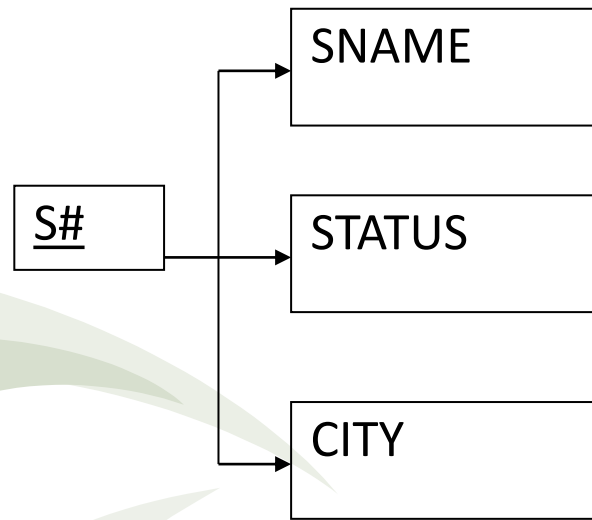
- Example from the rules

Let $R(A,B,C,G,H,I)$ and

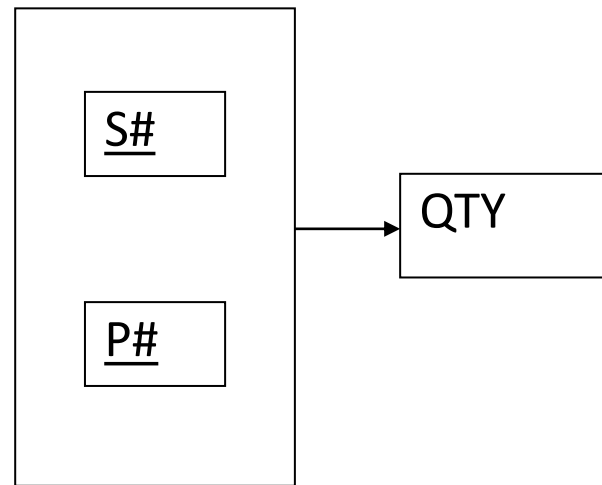
$FD = \{ A \rightarrow B, A \rightarrow C, CG \rightarrow H, CG \rightarrow I, B \rightarrow H \}$ then FD^+ are

- | | |
|-----------------------------|----------------------------|
| 1. $A \rightarrow B$ | there is in FD |
| 2. $A \rightarrow C$ | there is in FD |
| 3. $A \rightarrow BC$ | 1 union 2 |
| 4. $AG \rightarrow CG$ | 2 augmentation with G |
| 5. $CG \rightarrow H$ | there is in FD |
| 6. $CG \rightarrow I$ | there is in FD |
| 7. $CG \rightarrow HI$ | 5 union 6 |
| 8. $AG \rightarrow HI$ | 4 transitive 7 |
| 9. $AG \rightarrow BG$ | 1 augmentation with G |
| 10. $AG \rightarrow A$ | reflexivity |
| 11. $AG \rightarrow ABCGHI$ | 4 union 8 union 9 union 10 |
| 12. $B \rightarrow H$ | there is in FD etc. |

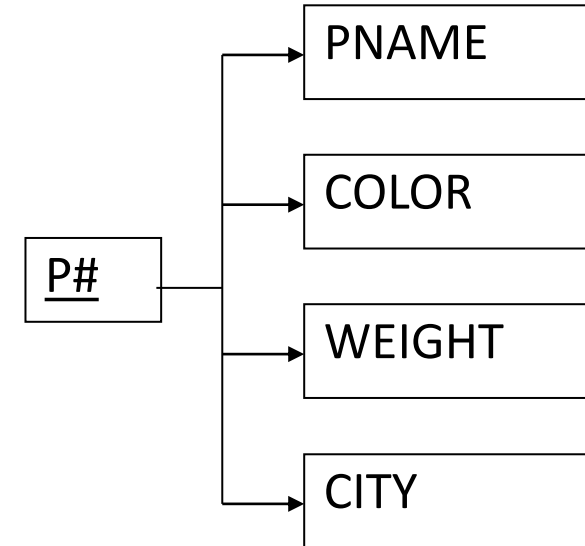
FD Diagram



Relation S



Relation SP



Relation P